NATIONAL ENGINEERING COLLEGE, K.R.NAGAR, KOVILPATTI – 628 503

(An Autonomous Institution, Affiliated to Anna University Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INNOVATIVE TEACHING METHODOLOGY

Academic Year: 2025–2026 Even Semester

COURSE CODE & COURSE NAME	**:**	**23CS24C - Object Oriented Programming using C++**
TOPIC CONDUCTED ON	:	**OOPS Concepts using C++**

ACTIVITY NAME: OOPS Marathon 2026

Date: 14.05.2026

Time: 02.00pm to 05.00pm

Venue: APJ & CVR labs

ACTIVITY DESCRIPTION:

The OOPS Marathon was conducted to strengthen students' understanding of Object-Oriented Programming concepts through real-time problem-solving activities and mini-project-based scenarios. Students were assigned various application-oriented scenarios to implement concepts including inheritance, polymorphism, function overloading, operator overloading, templates, exception handling, file handling, and STL containers. The activity encouraged students to design modular and reusable programs using C++ programming principles. Students actively participated in coding, debugging, and presenting their solutions, which enhanced their logical thinking, programming skills, and creativity. The marathon provided hands-on exposure to real-world software development practices and helped students gain confidence in applying OOPS concepts effectively in practical applications. To recognize exceptional performance and innovative problem-solving skills, appreciation gifts were awarded to the top three students who demonstrated extraordinary coding excellence and outstanding implementation during the marathon.

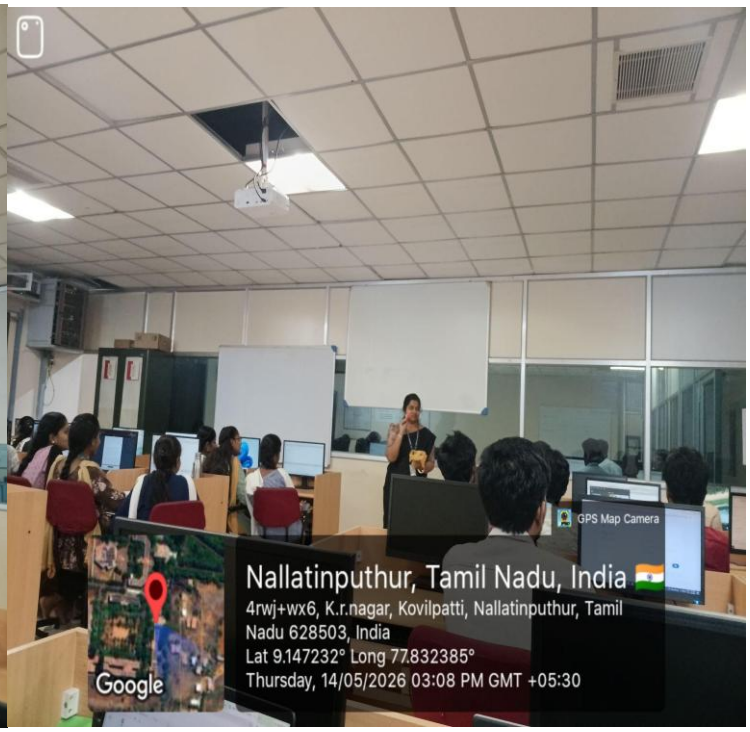
ACTIVITY OBJECTIVE:

- To provide practical understanding of Object-Oriented Programming concepts and advanced C++ programming techniques.
- To enhance students' problem-solving, coding, debugging, and software design skills through real-world programming applications.
- Students were grouped based on their learning proficiency levels as Slow Learners, Mid Learners, and Fast Learners.

- Each group was provided with programming challenges of appropriate complexity, enabling students to learn and perform at their skill level.
- A total of 76 students participated in the OOPS Marathon from a class strength of 129, with participants grouped into teams of three.

ACTIVITY SNAPSHOTS:











Content Available at:

https://drive.google.com/drive/folders/1qSUzT0I1Bv0wiDA3CEH1WG5zLywuc-oR?usp=sharing

Course Instructors:

Ms.R.Vazhan Arul Santhiya, AP/CSE

Ms.R.Navedha Evanjalin, AP/CSE

Ms.T.Rajeswari, AP/CSE

Head of the Department

Dean Academic

Principal

FEEDBACK OF STUDENTS

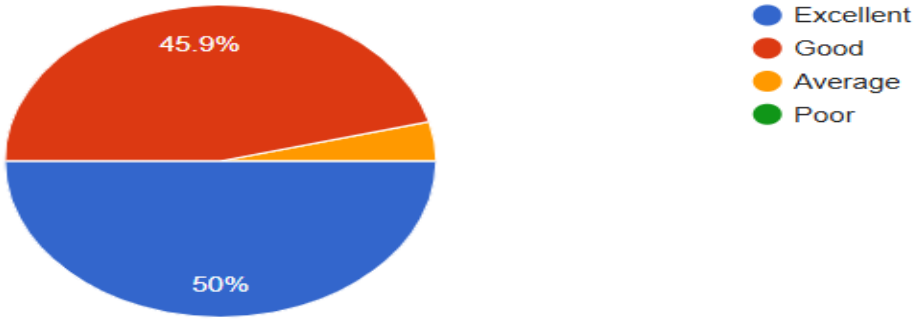
ACTIVITE LEARNING: OOPS Marathon 2026

[Response collected through Google Forms]

OOPS Marathon Responses.xlsx

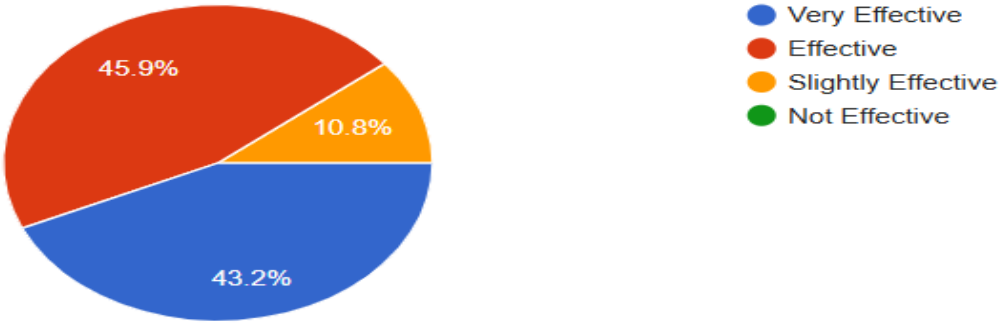
How would you rate your overall learning experience in the OOPS Marathon?

74 responses



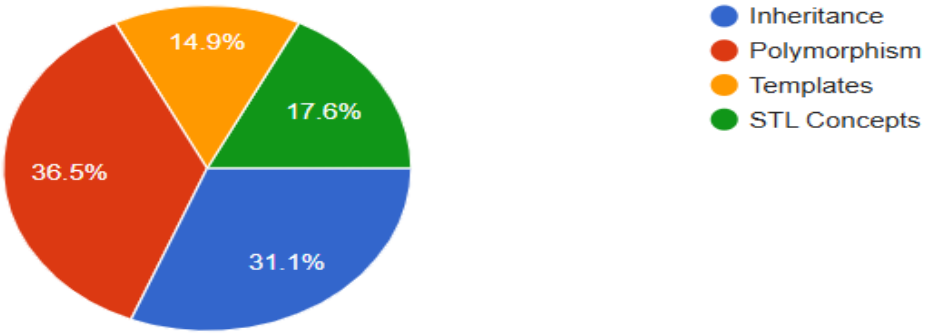
How effective were the problem statements in improving your programming skills?

74 responses



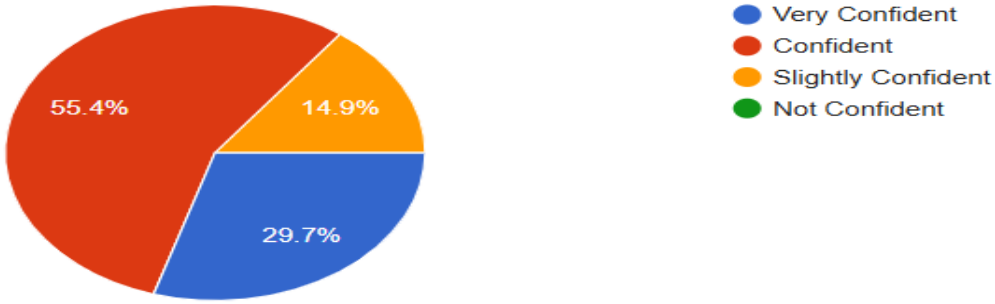
Which concept did you understand the best after completing the marathon?

74 responses



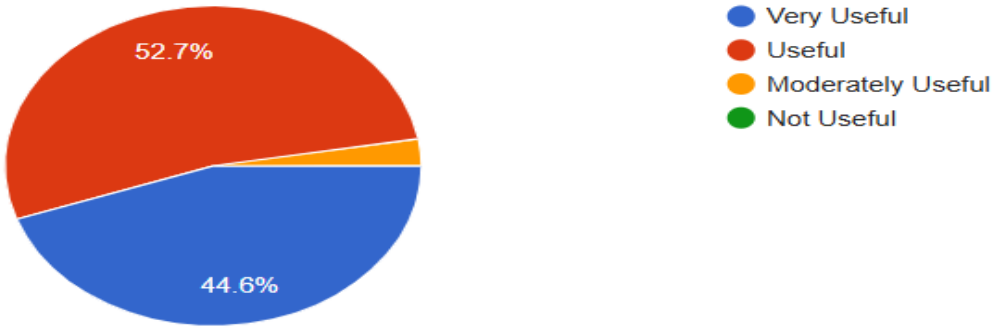
How confident are you in implementing exception handling after this activity?

74 responses



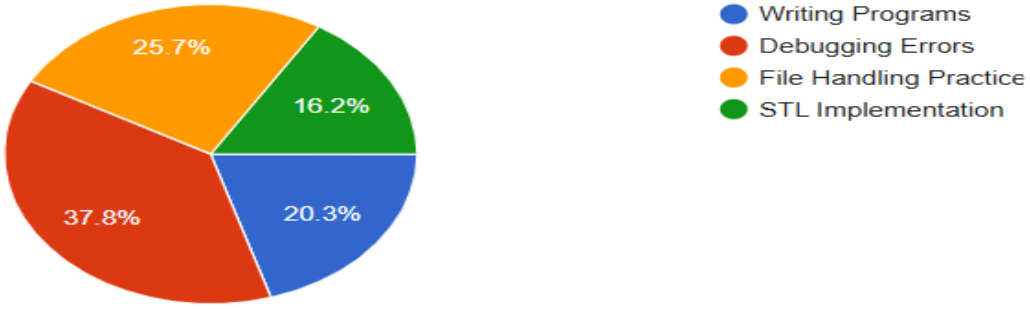
How useful were the real-time scenarios used in the questions?

74 responses



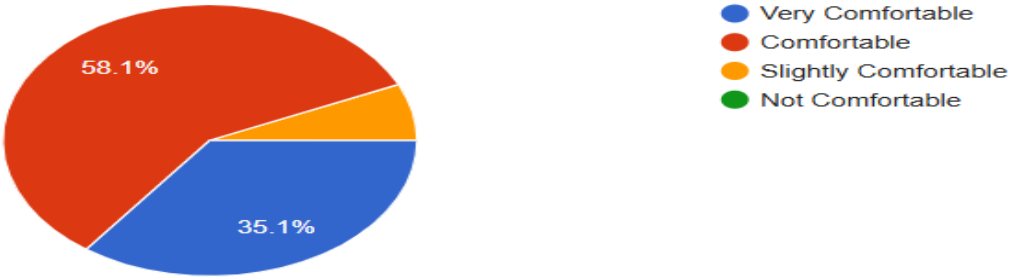
Which activity helped you the most during the marathon?

74 responses



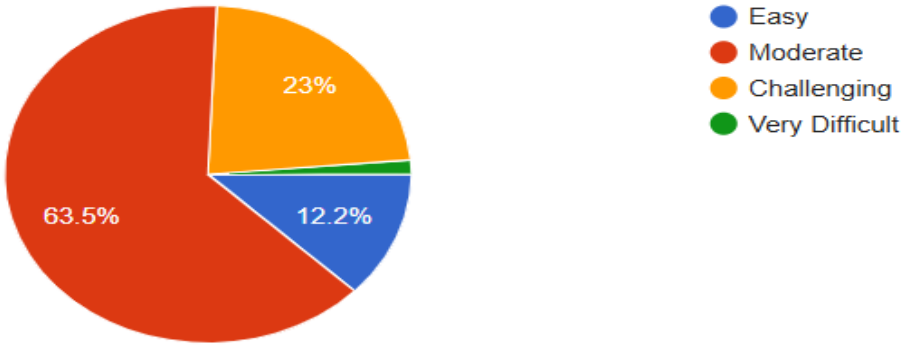
How comfortable are you now with implementing class hierarchies and polymorphism?

74 responses



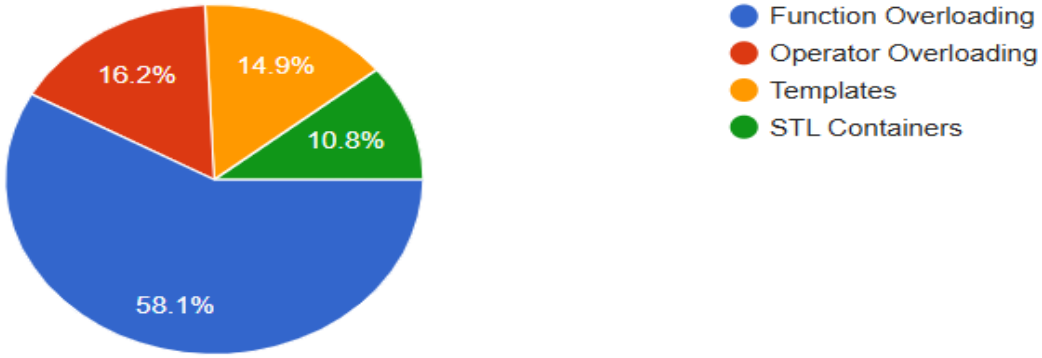
How would you rate the difficulty level of the marathon questions?

74 responses



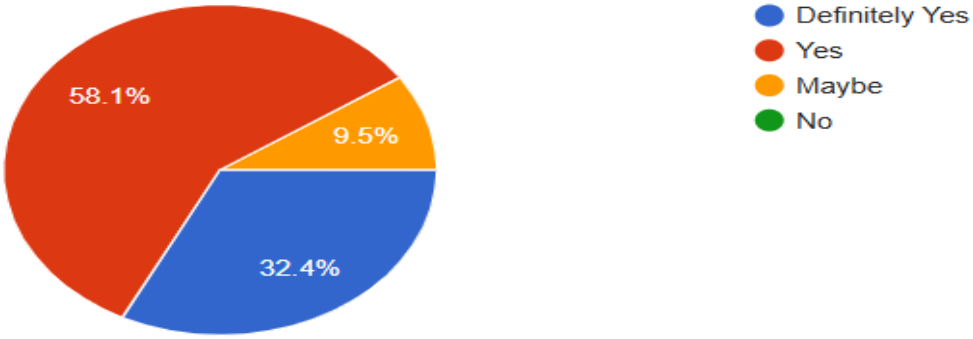
Which feature of C++ did you enjoy implementing the most?

74 responses



Would you like to participate in similar coding marathons in the future?

74 responses



What did you like most about the OOPS Marathon?
I liked the real-time application-based questions because they helped me understand how OOPS concepts are used in actual software development.
The marathon improved my coding confidence by making me implement concepts like inheritance, polymorphism, and STL practically.
I enjoyed solving different management system scenarios since each problem made programming more interesting and creative.
The activity helped me learn debugging and logical thinking skills while working on mini-project-style programs.
I liked the hands-on learning approach because it was more interactive and engaging than regular classroom exercises.

How did the OOPS Marathon help you in inferring the subject better?
The OOPS Marathon helped me understand how object-oriented concepts can be applied to real-world applications instead of learning only theory.
By solving different case-study-based problems, I was able to improve my programming logic and clearly understand concepts like inheritance, polymorphism, templates, and exception handling.
The activity made me confident in designing complete systems using C++ and helped me connect multiple OOPS concepts together in a single program.
Working on unique scenarios improved my analytical thinking and gave me a better understanding of how STL containers and file handling are used in practical applications.
The marathon provided hands-on experience in implementing object-oriented programming concepts, which helped me infer the subject in a more interactive and meaningful way.

29/8/24

HOD/CSE

DEAN ACADEMIC

PRINCIPAL 12/9